

| Question |     |      | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Marks available |            |     |       | Maths | Prac |
|----------|-----|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|------------|-----|-------|-------|------|
|          |     |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | AO1             | AO2        | AO3 | Total |       |      |
| 7        | (a) |      | [It has] magnitude (accept size) and direction                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 1               |            |     | 1     |       |      |
|          | (b) | (i)  | Total momentum before collision = $30\,000 + 15\,000$<br>[= $45\,000\text{ kg m s}^{-1}$ ] (1)<br>Total momentum after collision = $27\,000 + 18\,000$<br>[= $45\,000\text{ kg m s}^{-1}$ ] (1) Ignore units<br>Deduct 1 mark for powers of 10 slip<br>Award 1 mark only for -<br>Momentum is not lost [in collision] or momentum before [collision] is the same as momentum after [collision] / momentum is conserved (1) Don't accept: they are the same<br><b>Alternative:</b><br>Loss in momentum of A = $12\,000\text{ kg m s}^{-1}$ (1)<br>Gain in momentum of B = $12\,000\text{ kg m s}^{-1}$ (1)<br>Hence the gain in momentum of B = loss in momentum of A (1) | 1               | 1<br><br>1 |     | 3     | 2     |      |
|          |     | (ii) | Attempt at using $p_A + p_B = (m_A + m_B)v$ (1)<br>Correct substitution e.g. $45\,000\text{ ecf } = 25\,000v$ (1) Award 2 marks if this seen.<br>$v = 1.8\text{ m s}^{-1}$ (1)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 1               | 1<br><br>1 |     | 3     | 3     |      |
|          | (c) | (i)  | A body's <u>rate of</u> / change per second (reference to time) change of momentum (1) is proportional to [accept 'equal to'] the [resultant] force acting on it (1) [and is in the direction of this force]<br><b>Alternative:</b><br>Formula stated (1) with all terms defined (1)                                                                                                                                                                                                                                                                                                                                                                                     | 2               |            |     | 2     |       |      |
|          |     | (ii) | Time for collision = $0.2\text{ s}$ (1) accept $(0.5 - 0.3)$<br>$F = \frac{-12\,000}{0.2}$ (1)<br>$F = -60\,000\text{ [N]}$ (1) <b>ecf</b> on powers of 10 slip                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                 | 1<br><br>1 | 1   | 3     | 3     |      |